

Rings of Polynomials

Algebraic number theory concerns itself with the solutions of systems of polynomial equations with integer coefficients.

The ring $\mathbb{Z}$ of integers is a *domain*.

In other words, it is a commutative ring with identity Z such that, whenever $ab = 0$ in Z for some a, b in Z , at least one of a or b has to be 0. Note that this is the same as the assertion that if $a \neq 0$ is in Z , then the map $Z \rightarrow Z$ given by multiplication by a is one-to-one.

Instead of merely studying polynomials with integer coefficients, it is convenient to study polynomials with coefficients in a domain.

Units and fields

An element u in a commutative ring R with identity is called a *unit* if multiplication by u gives an bijection $R \rightarrow R$.

Exercise: Show that this is the same as the assertion that the map $R \rightarrow R$ given by multiplication by u is onto. (Hint: Note that 1 is in the image.)

When *every* non-zero element a in a commutative ring Q is a unit, we say that Q is a *field*.

We can mimic the construction of rationals from integers as follows. Given a domain Z , we can form the *field of fractions* $Q(Z)$ of Z :

- An element of $Q(Z)$ is an equivalence class of pairs (a, b) with $a, b \in Z$ and $b \neq 0$.
- Pairs (a, b) and (c, d) are said to be equivalent if $ad = bc$ in Z . One checks that this is an equivalence relation.
- The additive identity in $Q(Z)$ is the equivalence class of pairs $(0, b)$ with $b \neq 0$.
- The multiplicative identity in $Q(Z)$ is the equivalence class of pairs (a, a) with $a \neq 0$.
- The sum of (a, b) with (c, d) is the equivalence class of $(ad + bc, bd)$. Note that $b \neq 0$ and $d \neq 0$ implies $bd \neq 0$.
- The product of (a, b) with (c, d) is the equivalence class of (ac, bd) .

Exercise: Check that $Q(Z)$ is a field. (Hint: A non-zero element is an equivalence class containing a pair (a, b) with $ab \neq 0$ and so (b, a) is its multiplicative inverse.)

Conversely, if Q is a field and $Z \subset Q$ is a subring (containing 1), then it is clear that Z is a domain and $Q(Z)$ can be seen as a subfield of Q via the homomorphism $(a, b) \mapsto a/b$.

Note that a field Q can contain more than one domain Z . For example, given any positive integer n the collection of rational numbers of the form a/n^k is the domain $\mathbb{Z}[1/n]$ inside the field $\mathbb{Q}$ of rational numbers.

Exercise: Give an example of a proper sub-domain of $\mathbb{Q}$ which is *not* of the form $\mathbb{Z}[1/n]$.

We will often use arguments involving $Q(Z)$ to study Z and vice versa.

Polynomial ring

Given R a commutative ring with identity, we can think of the polynomial ring $R[X]$ as the collection of expressions of the form $a_0 + a_1X + \cdots + a_nX^n$ where a_i 's are elements of R and n is a non-negative integer. Addition and multiplication are defined in the usual way.

Alternatively, we can think of $R[X] = \oplus_{i=0}^{\infty} R$ as the collection of sequences which are “eventually zero”:

$$R[X] = \{(a_i)_{i \geq 0} : \forall i, a_i \in R \text{ and } \exists n, \forall i, i > n \Rightarrow a_i = 0\}$$

- The sum of $\mathbf{a} = (a_i)$ and $\mathbf{b} = (b_i)$ is defined as $\mathbf{a} + \mathbf{b} = (c_i)$ where $c_i = a_i + b_i$.
- The product of $\mathbf{a} = (a_i)$ and $\mathbf{b} = (b_i)$ is defined as $\mathbf{a} \cdot \mathbf{b} = (c_i)$ where $c_i = \sum_{j=0}^i a_j b_{i-j}$.
- A “constant” $a \in R$ is represented in this by the sequence (a_i) with $a_0 = a$ and $a_i = 0$ for $i > 0$. By abuse of notation we denote this element of $R[X]$ by a .
- The “variable” X is represented in this ring as the sequence $X = (\ell_i)$ with $\ell_1 = 1$ and $\ell_i = 0$ for $i \neq 1$.

Note that multiplication by X is the *right shift operator* which takes the sequence (a_i) to the sequence (b_i) where $b_0 = 0$ and $b_{i+1} = a_i$ for all i .

Given a non-zero polynomial (a_i) , the largest d such that $a_d \neq 0$ is called the *degree* of the polynomial. We write (a_i) more conventionally as $a_0 + a_1X + \cdots + a_dX^d$ where d is its degree and $a_d \neq 0$.

Exercise: If $\mathbf{a}$ is a non-zero polynomial of degree d and $\mathbf{b}$ is a non-zero polynomial of degree e , then their product $\mathbf{a} \cdot \mathbf{b} = (c_i)$ satisfies $c_{d+e} = a_d b_e$ and $c_k = 0$ for $k > d + e$. Further, note that when R is a domain then c_{d+e} is non-zero.

The degree of the product of non-zero polynomials with coefficients in a domain is the sum of their degrees.

In particular, we note:

If Z is a domain then the polynomial ring $Z[X]$ is a domain.

It is sometimes convenient to define the degree of the 0 polynomial as $-\infty$.

- The degree of the sum of two polynomials is at most the maximum of the degree of each polynomial.
- The degree of the product of two polynomials is the sum of the degrees of each polynomial. (We use the convention that $-\infty + a = -\infty$.)

If $\mathbf{u}$ is a unit in $Z[X]$, then there is a $\mathbf{v}$ in $Z[X]$ such that $\mathbf{u} \cdot \mathbf{v} = 1$. By computing degrees on both sides, we see that the degree of $\mathbf{u}$ and $\mathbf{v}$ is 0; i.e. $\mathbf{u} = u$ and $\mathbf{v} = v$ are units in Z .

A polynomial with coefficients in a domain Z is a unit if and only if it is a constant and is a unit in Z .

In passing, we note that the degree of polynomials is similar in terms of algebraic properties to $-v_p$ on rational numbers (v_p is the p -adic valuation for a prime p)!

Divisors and irreducible elements

Given a an element of Z , consider the map $Z \rightarrow Z$ given by multiplication by a . An element b in the image of this map is called a *multiple* of a and a is called a *divisor* of b . Both of these are equivalent to the assertion that there is an element c in Z such that $b = a \cdot c$; since Z is a domain, the element c is uniquely determined by a and b .

Note the units are trivially divisors of all elements. Conversely, all elements are multiples of units.

An element a in Z is called *irreducible* if, whenever b is a non-unit divisor of a , we have that a is a divisor of b . Writing $a = b \cdot c$ and $b = a \cdot d$ we see that $a \cdot b = (a \cdot b)c \cdot d$, from which it follows that $1 = c \cdot d$ so that c and d are units.

Thus, we can also say that a non-unit a is irreducible if every non-unit divisor of a is a unit times a .

The set of divisors. Given a non-unit a in a domain Z , we define the set $\text{Div}(a)$ to be the set of divisors of a .

It is clear that $\text{Div}(0) = Z$.

Note that if p is irreducible in Z , then $\text{Div}(p)$ consists of units and unit multiples of p .

Given two elements a and b in Z , we use $\text{Div}(a, b) = \text{Div}(a) \cap \text{Div}(b)$ to denote the set of common divisors of a and b in Z .

We see that $\text{Div}(a, 0) = \text{Div}(a)$.

We also see that if p is irreducible in Z , then:

- If p is a divisor of b then $\text{Div}(p, b) = \text{Div}(b)$.
- If p is not a divisor of b , then $\text{Div}(p, b)$ consists only of units in Z .

Factorisation. One sees easily that if $Z = \mathbb{Z}$ is the ring of integers, then irreducible elements are of the form $\pm p$ where $p \in \mathbb{N}$ is a prime number.

Mimicking what we know about integers, we say that an element a in Z is *factored* into irreducible elements if we have an expression $a = p_1 \cdots p_n$ where $p_1, \dots, p_n$ are irreducible elements of Z .

We now examine a condition under which such a factorisation is possible.

Suppose that there is some positive integer $\partial(a)$ associated with every non-zero element a of Z such that:

- $\partial(a \cdot b) = \partial(a) + \partial(b)$
- $\partial(a) = 0$ if and only if a is a unit.
- $\partial(a + b) \leq \max\{\partial(a), \partial(b)\}$.

For complete-ness, we also define $\partial(0) = -\infty$.

An important example is $Z = Q[X]$ for a field Q , where the degree of a polynomial is such a ∂ .

Exercise: Show that $a \in Z$ is irreducible if and only if $\partial(a) > 0$ and for every divisor b of a we have $\partial(b) = 0$ or $\partial(b) = \partial(a)$.

We can now use downward induction to prove:

Exercise: For every $a \in Z$ with $\partial(a) > 0$, there is an irreducible element $p \in Z$ such that p is a divisor of a .

Another application of induction can then be used to prove:

Exercise: For every $a \in Z$ with $\partial(a) > 0$, there is an expression $a = p_1 \cdots p_k$ where p_i are irreducible elements of Z .

In particular, we note that we have such a factorisation in the case $Z = Q[X]$, where Q is a field.

Unique factorisation domain (UFD). We say that a factorisation is unique upto permutations and units if, given another expression $a = q_1 \cdots q_m$ where $q_1, \dots, q_m$ are irreducible, we have $m = n$ and a permutation σ of $\{1, \dots, n\}$ such that q_i is a unit multiple of $p_{\sigma(i)}$.

A domain Z is a *unique factorisation domain* (or UFD) if every element of Z has a factorisation which is unique upto permutations and units.

Exercise: Consider the subring $\mathbb{Z}[\sqrt{-5}] = \{a + b\sqrt{-5} : a, b \in \mathbb{Z}\}$ of the field $\mathbb{C}$ of complex numbers. Note that

$$(1 + \sqrt{-5})(1 - \sqrt{-5}) = 2 \cdot 3 = 6$$

Check that these factorisations of 6 are not equivalent upto permutation and units in $\mathbb{Z}[\sqrt{-5}]$. (Hint: Show that the only units in $\mathbb{Z}[\sqrt{-5}]$ are ± 1 .)

Thus, there are domains that are *not* UFDs. Thus we can look for conditions applicable to polynomials that will give us a UFD.

Long division

Given a non-zero polynomial $\mathbf{a}$ of degree d and a non-zero polynomial $\mathbf{b}$ of degree e with $d \geq e \geq 0$, we note that the polynomial $\mathbf{c} = b_e \mathbf{a} - a_d X^{d-e} \mathbf{b}$ has degree

less than d . In the usual notation, this can be written as

$$\begin{aligned} c_0 + c_1X + \cdots + c_{d_1}X^{d_1} \\ = b_e (a_0 + a_1X + \cdots + a_dX^d) \\ - a_dX^{d-e} (b_0 + b_1X + \cdots + b_eX^e) \end{aligned}$$

which makes it clear that $d_1 < d$. If the degree d_1 of $\mathbf{c}$ satisfies $d_1 \geq e$, we can repeat this process.

Exercise: Given polynomials $\mathbf{a}$ and $\mathbf{b} \neq 0$ of degrees d and e respectively such that $d \geq e \geq 0$. Show that we can find polynomials $\mathbf{r}$ and $\mathbf{q}$ such that

$$\mathbf{r} = b_e^{d-e} \mathbf{a} - \mathbf{q}\mathbf{b}$$

where $\mathbf{r}$ has degree f with $f < e$. (Note that $\mathbf{r}$ may be 0.)

A polynomial $\mathbf{b}$ in $Z[X]$ is said to be *monic* if its leading (highest degree) coefficient b_e is a unit in the coefficient ring Z . (Note that a monic polynomial is, by definition, non-zero!)

The equation $\mathbf{r} = b_e^k \mathbf{a} - \mathbf{q} \cdot \mathbf{b}$ can be divided by b_e^k when b_e is a unit.

Given polynomials $\mathbf{a}$ and $\mathbf{b}$ such that $\mathbf{b}$ is monic, we can find polynomials $\mathbf{r}$ and $\mathbf{q}$ such that $\mathbf{r} = \mathbf{a} - \mathbf{q} \cdot \mathbf{b}$ where $\mathbf{r}$ has degree less than the degree of $\mathbf{b}$.

This is the form in which long division is often used.

In particular, given a field Q , any non-zero polynomial in $Q[X]$ is monic. Thus, long division in the above form applies to polynomials with coefficients in a field.

GCD algorithm. In this section we work with polynomials with coefficients in a field Q .

Since we have long division, we can carry out the following recursive algorithm to calculate a polynomial $\gcd(\mathbf{a}, \mathbf{b})$.

1. If the degree of $\mathbf{b}$ is greater than the degree of $\mathbf{a}$, we define $\gcd(\mathbf{a}, \mathbf{b}) = \gcd(\mathbf{b}, \mathbf{a})$.
2. We define $\gcd(\mathbf{a}, 0) = \mathbf{a}$.
3. If $\mathbf{r} = \mathbf{a} - \mathbf{q} \cdot \mathbf{b}$ by long division, then we define $\gcd(\mathbf{a}, \mathbf{b}) = \gcd(\mathbf{b}, \mathbf{r})$ and go back to step (2) to calculate this.

In steps (2) and (3), we note that, by step (1), the degree of $\mathbf{a}$ is at least the degree of $\mathbf{b}$. Thus, long division is possible. Moreover, after step (3), we have reduced the computation to polynomials of smaller degree. Thus, by downward induction, the process terminates giving us an algorithm which always exits via step (2).

Moreover, we see that the degree of $\gcd(\mathbf{a}, \mathbf{b})$ is at most the maximum of the degrees of $\mathbf{a}$ and $\mathbf{b}$.

We note that $\mathbf{r}$ is a linear combination of $\mathbf{a}$ and $\mathbf{b}$; i.e. it is of the form $\mathbf{a} \cdot \mathbf{p} + \mathbf{b} \cdot \mathbf{q}$ for suitable polynomials $\mathbf{p}$ and $\mathbf{q}$. A linear combination of linear combinations of $\mathbf{a}$ and $\mathbf{b}$ is also a linear combination of $\mathbf{a}$ and $\mathbf{b}$. Thus, we see that at each stage we are calculating $\gcd(\mathbf{a}', \mathbf{b}')$ where $\mathbf{a}'$ and $\mathbf{b}'$ are linear combinations of $\mathbf{a}$ and $\mathbf{b}$. Since we exit at step (2), we see that

$$\gcd(\mathbf{a}, \mathbf{b}) = \mathbf{a} \cdot \mathbf{p} + \mathbf{b} \cdot \mathbf{q}$$

for suitable polynomials $\mathbf{p}$ and $\mathbf{q}$. Given that the degree of the left-hand side is at most the maximum of the degrees of $\mathbf{a}$ and $\mathbf{b}$, we apply long division to show that the degree of $\mathbf{p}$ is less than the degree of $\mathbf{b}$ and the degree of $\mathbf{q}$ is less than the degree of $\mathbf{a}$ unless one of $\mathbf{a}$ or $\mathbf{b}$ is 0.

Greatest Common Divisor and gcd. Recall that for an element a in a domain Z , we introduced the set $\text{Div}(a)$ as the set of divisors of a and defined $\text{Div}(a, b) = \text{Div}(a) \cap \text{Div}(b)$.

Given $c = a - q \cdot b$, we see that any d that divides a and b divides c ; conversely, any d that divides b and c divides a .

$$\text{Div}(a, b) = \text{Div}(b, a - q \cdot b)$$

We now specialise to the case $Z = Q[X]$ where Q is a field.

We apply the above identity to step (3) in GCD algorithm above to obtain:

$$\text{Div}(\mathbf{a}, \mathbf{b}) = \text{Div}(\mathbf{b}, \mathbf{r})$$

Finally, in step (1) we note that $\text{Div}(\mathbf{a}, 0) = \text{Div}(\mathbf{a})$.

Thus, we see that the GCD algorithm can be used to calculate common divisors:

$$\text{Div}(\mathbf{a}, \mathbf{b}) = \text{Div}(\gcd(\mathbf{a}, \mathbf{b}))$$

If $\mathbf{a}$ is non-zero, we note that divisor of $\mathbf{a}$ has degree at most the degree of $\mathbf{a}$; in other words, $\mathbf{a}$ is the element of largest degree in $\text{Div}(\mathbf{a}, 0)$ in this case.

We deduce that $\gcd(\mathbf{a}, \mathbf{b})$ is an element of largest degree in $\text{Div}(\mathbf{a}, \mathbf{b})$ when at least one of these polynomials is non-zero. Hence, it is called the *greatest common divisor* of $\mathbf{a}$ and $\mathbf{b}$.

In particular, combining the above we have:

Exercise: If $\mathbf{p}$ is an irreducible element in $Q[X]$ which is not a divisor of $\mathbf{a}$, then there are polynomials $\mathbf{r}$ and $\mathbf{s}$ such that

$$\mathbf{r}\mathbf{p} + \mathbf{s}\mathbf{a} = 1$$

Note that this applies, in particular, to the case where the degree of $\mathbf{a}$ is less than the degree of the irreducible polynomial $\mathbf{p}$.

Primes

Suppose a prime number p divides $a \cdot b$ in $\mathbb{Z}$, and p does not divide a . By the Euclidean GCD algorithm, there are integers r and s such that $1 = r \cdot p + s \cdot a$. It follows that $b = r \cdot p \cdot b + s \cdot a \cdot b$. Since p divides $a \cdot b$, we see that p divides the right-hand side, and so p divides b .

An element p in a domain Z is said to be *prime* if, whenever p is a divisor of $a \cdot b$ and p is not a divisor of a , it is a divisor of b .

Exercise: Show that if p is a prime element of a domain Z , then it is irreducible in Z .

Note that the converse is not, in general, true! However, a special case is when $Z = Q[X]$ where Q is a field.

Exercise: Show that if $\mathbf{p}$ is an irreducible element of $Q[X]$, then it is a prime in the sense defined above. (Hint: Use the GCD expression.)

Unique factorisation in $Q[X]$. We have seen that every non-unit element in $Q[X]$ can be factorised into irreducible elements.

To prove essential uniqueness of factorisation in a domain Z , we need to know that if q is irreducible and divides $p_1 \cdots p_k$, then q divides one of the p_i . By the irreducibility of p_i it would follow that q and p_i are unit multiples of each other.

Exercise: Show that a domain Z is a UFD if:

- Every non-unit element can be written as a product of irreducible elements.
- Every irreducible element is prime.

We conclude that $Q[X]$ is a UFD, when Q is a field.

More examples of UFD's

If Z is a domain and p is a prime in Z , then consider the quotient ring $Z/\langle p \rangle$. If a and b are elements of Z such that $a \cdot b$ becomes 0 in $Z/\langle p \rangle$, then p divides $a \cdot b$ in Z . By the primality of p , it follows that either p divides a or p divides b .

Given a domain Z by a prime p , the quotient $Z/\langle p \rangle$ is also a domain.

Content of a polynomial. We now consider polynomials in $Z[X]$ where Z is a UFD.

Note that in a UFD, since every element can be written as a product of irreducible elements, we can take “common irreducible divisors” to form the “greatest common divisor” of any pair of elements of the domain. By induction, we can find the greatest common divisor of any finite set of elements of the domain. In other words:

Exercise: Given a finite collection of elements $a_1, \dots, a_r$ of a UFD Z , we can find an element g in Z such that $a_i = g \cdot b_i$ in Z for all i such that if d is a common divisor of b_i for all i , then b is a unit in Z .

We call g a greatest common divisor of $a_1, \dots, a_r$. It is well-determined upto a unit in Z , so sometimes, by abuse of terminology, we call it *the* greatest common divisor.

Given $\mathbf{a} \in Z[X]$, the *content* of $\mathbf{a}$ is defined as the greatest common divisor of the coefficients of a . Note that this is determined up to a unit by the irreducible (prime) factors of the coefficients of $\mathbf{a}$.

Suppose that $\mathbf{a} = \mathbf{b} \cdot \mathbf{c}$ in $Z[X]$ and suppose that p is a prime in Z that divides the content of $\mathbf{a}$.

It follows that $\mathbf{b} \cdot \mathbf{c}$ becomes 0 in the domain $(Z/\langle p \rangle)[X]$. This means that either $\mathbf{b}$ or $\mathbf{c}$ becomes 0 in this domain. In turn, that means the p divides the content of either $\mathbf{b}$ or $\mathbf{c}$. Since an element of Z is determined up to a unit by its prime factors:

When Z is a UFD, the content of a product $\mathbf{a} = \mathbf{b} \cdot \mathbf{c}$ is the product of the contents of $\mathbf{b}$ and $\mathbf{c}$ upto a unit.

Factorisation in $Z[X]$ when Z is as UFD. We denote the field of fraction $Q(Z)$ by Q for ease of notation.

We have already seen that $Q[X]$ is a UFD.

Given a polynomial $\mathbf{a}$ in $Z[X]$ with unit content. Suppose there is a factorisation $\mathbf{a} = \mathbf{b} \cdot \mathbf{c}$ in $Q[X]$.

By “clearing denominators” we can write this as $d\mathbf{a} = \mathbf{b}' \cdot \mathbf{c}'$ in $Z[X]$, where $\mathbf{b}$ is a Q -multiple of $\mathbf{b}'$ and $\mathbf{c}$ is a Q -multiple of $\mathbf{c}'$, and d is some element of $\mathbf{Z}$. The content of $\mathbf{b}' \cdot \mathbf{c}'$ is the product of their contents, and the latter is a unit multiple of d . It follows that $d = d_1 \cdot d_2$ where d_1 divides the content of $\mathbf{b}'$ and d_2 divides the content of $\mathbf{c}'$.

Thus, we see that we have $\mathbf{a} = \mathbf{b}_1 \cdot \mathbf{c}_1$ where the latter two polynomials are in $Z[X]$.

If Z is a UFD and Q its field of fractions, then any factorisation $\mathbf{a} = \mathbf{b} \cdot \mathbf{c}$ in $Q[X]$ of a polynomial $\mathbf{a}$ in $Z[X]$ yields a factorisation $\mathbf{a} = \mathbf{b}_1 \cdot \mathbf{c}_1$ in $Z[X]$ where $\mathbf{b}$ and $\mathbf{c}$ are Q -multiples of $\mathbf{b}_1$ and $\mathbf{c}_1$ respectively.

We easily conclude that the factorisation into irreducibles (primes) of a polynomial $\mathbf{a} \in Z[X]$ in the domain $Q[X]$ yields a factorisation into irreducibles in $Z[X]$.

If Z is a UFD, then so is $Z[X]$.

For clarity, it is worth noting that there are two types of primes in $Z[X]$.

- The primes in Z are also primes in $Z[X]$.
- A polynomial $\mathbf{a} \in Z[X]$ which is prime in $Q[X]$ and has unit content in $Z[X]$.

This allows us to show that the following important examples of polynomial rings are UFD's.

The rings $R[X_1, \dots, X_n]$ are UFD's for R as any of the rings $\mathbb{Z}$, $\mathbb{Z}_p$ or any field.